

Mining Application – Docker Environment Setup Guide

Separate Docker environment using the same technology approach as RAMS

1. Folder Structure

```
C:\Users\ADMIN\Veeresh\Projects\  
■■■ RAMS\  
■■■ Mining\  
    ■■■ docker-compose.yml  
    ■■■ database\  
        ■■■ Dockerfile  
        ■■■ init\  
            ■■■ 01-init.sql  
    ■■■ geoserver\  
    ■■■ data\
```

2. Create Folders

```
cd C:\Users\ADMIN\Veeresh\Projects\Mining
```

```
New-Item -ItemType Directory -Force database  
New-Item -ItemType Directory -Force database\init  
New-Item -ItemType Directory -Force geoserver  
New-Item -ItemType Directory -Force data
```

3. PostgreSQL/PostGIS + pgRouting Dockerfile

```
Create: database\Dockerfile  
FROM postgis/postgis:17-3.5
```

```
RUN apt-get update && \  
    apt-get install -y postgresql-17-pgrouting && \  
    rm -rf /var/lib/apt/lists/*
```

4. Database Initialization

```
Create: database\init\01-init.sql  
CREATE EXTENSION IF NOT EXISTS postgis;  
CREATE EXTENSION IF NOT EXISTS postgis_raster;  
CREATE EXTENSION IF NOT EXISTS pgrouting;  
  
CREATE SCHEMA IF NOT EXISTS mining;  
CREATE SCHEMA IF NOT EXISTS gis;  
CREATE SCHEMA IF NOT EXISTS audit;
```

5. docker-compose.yml

```
services:
```

```
  mining_postgis:  
    build:  
      context: ./database  
      dockerfile: Dockerfile  
    container_name: mining_postgis  
    environment:  
      POSTGRES_DB: mining_db  
      POSTGRES_USER: mining_user  
      POSTGRES_PASSWORD: mining_password  
    ports:  
      - "5434:5432"  
    volumes:  
      - mining_postgis_data:/var/lib/postgresql/data  
      - ./database/init:/docker-entrypoint-initdb.d  
    networks:  
      - mining_network  
    restart: unless-stopped  
  
  mining_geoserver:  
    image: kartoza/geoserver:2.28.0  
    container_name: mining_geoserver  
    ports:  
      - "8082:8080"  
    volumes:
```

```

    - mining_geoserver_data:/opt/geoserver/data_dir
networks:
  - mining_network
restart: unless-stopped
depends_on:
  - mining_postgis

networks:
  mining_network:
    name: mining_network
    driver: bridge

volumes:
  mining_postgis_data:
    name: mining_postgis_data

  mining_geoserver_data:
    name: mining_geoserver_data

```

Explicit volume names prevent duplicated names such as mining_mining_postgis_data.

6. Check Docker Desktop

```

docker version
docker ps

```

Docker Desktop must be running. Do not stop or remove the existing RAMS containers.

7. Validate Compose

```

cd C:\Users\ADMIN\Veeresh\Projects\Mining
docker compose config

```

This validates the Compose configuration without starting the services.

8. Build

```

docker compose build

```

For a completely fresh build: `docker compose build --no-cache`

9. Start Mining

```

docker compose up -d

```

-d means detached mode: containers run in the background and PowerShell returns to the prompt.

10. Check Containers

```

docker ps

```

Expected Mining containers: mining_postgis and mining_geoserver. RAMS containers may also appear.

11. Check Logs

```

docker logs mining_postgis --tail 50
docker logs mining_geoserver --tail 50

```

For continuous logs use `docker logs -f mining_postgis` or `docker logs -f mining_geoserver`. Ctrl+C only stops log viewing.

12. Test PostGIS and pgRouting

```

docker exec -it mining_postgis psql -U mining_user -d mining_db

```

```

Inside PostgreSQL:
SELECT PostGIS_Version();
SELECT pgr_version();
\dn
\q

```

Expected schemas include audit, gis, mining and public.

13. Open GeoServer

http://localhost:8082/geoserver

14. Ports

Service	RAMS	Mining
PostgreSQL	5432	5434
GeoServer	8081	8082
Backend	8000	8001 later
Frontend	5173	5174 later

15. Safe Daily Commands

Start

```
cd C:\Users\ADMIN\Veeresh\Projects\Mining
docker compose up -d
```

Check

```
docker ps
```

Stop temporarily

```
docker compose stop
```

Start stopped containers

```
docker compose start
```

Remove containers/network but keep named volumes

```
docker compose down
```

Rebuild after Docker configuration changes

```
docker compose up -d --build
```

16. Important Warning

```
docker compose down -v
```

Do not use this casually. The `-v` option removes named volumes and can delete the Mining database and GeoServer data. Use it only when intentionally resetting the complete Mining environment.

17. Final Architecture

MINING

■

■■■ mining_postgis
■ ■■■ PostgreSQL 17
■ ■■■ PostGIS
■ ■■■ pgRouting

■

■■■ mining_geoserver
■ ■■■ GeoServer 2.28.0

■

■■■ mining_postgis_data
■■■ mining_geoserver_data
■■■ mining_network

Later:

React + MapLibre
Mining Backend
AI Agent
RBAC / Login
Dashboards
Spatial Analysis
pgRouting
Alerts

18. Quick Reference

```
# Start
cd C:\Users\ADMIN\Veeresh\Projects\Mining
docker compose up -d

# Check
docker ps

# Stop safely
docker compose stop

# Start again
docker compose start

# Remove containers but keep data
docker compose down
```

This guide is the standard reference for the separate Mining Docker environment. RAMS remains a separate Docker project, database, network and set of containers.